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MILITARY SPECIFICATION

DIFFERENTIAL PRESSURE TRANSDUCER EQUIPMENT (ELECTRICAL) (NAVAL SHIPBOARD USE)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for electrical output differential pressure transducer equipment for Naval ships, but does not include the readout or display.

1.2 Classification.

1.2.1 Designations. Designations shall be given for the following as specified (see 6.2.1) and listed in the format below:

PDT	D	SG	ST	15	WCD	1500	4
Type (see 1.2.1.1)	Sensing technique (see 1.2.1.2)	Transduction principle (see 1.2.1.3)	Application (see 1.2.1.4)	Range (see 1.2.1.5)	Units (see 1.2.1.6)	Pressure rating (see 1.2.1.7)	Power supply (see 1.2.1.8)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.1 Type. Transducers shall be designated by the three-letter symbol as follows:

PDT - Pressure, differential, transducer.

1.2.1.2 Sensing technique (force summing devices). The sensing technique shall be designated by the corresponding letter symbols and are limited to the following:

D - Diaphragm.
B - Bellows.
T - Straight tube.

1.2.1.3 Transduction principle. The transduction principles shall be designated by the corresponding symbols and are limited to the following:

SG - Strain gauge.
VR - Variable reluctance.
DT - Differential transformer.
CP - Capacitance

1.2.1.4 Application. The fluid pressure to be measured shall be indicated as follows:

ST - Steam.
OS - Oil.
SW - Seawater.
FG - Flue gas and ammonia.
OX - Oxygen.
GW - Other gases, fresh water and condensate.

1.2.1.5 Range. The full scale differential pressure range in pounds per square inch (lb/in^2) differential, or inches of water column differential shall be designated by the numerical value of the upper range limit (see 3.4.3).

1.2.1.6 Units. The units shall be designated by the corresponding letter symbols and are limited to the following:

PSID - Pounds per square inch, differential.
WCD - Inches of water column, differential.

1.2.1.7 Pressure rating. The pressure rating shall be designated by its numerical value and is limited to the following as specified (see 6.2.1):

Pressure rating (lb/in^2)
150
600
1500
3000
6000

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1.2.1.8 Power supply. Wiring of the power supply shall be one of the following (see 3.4.4):

2 - Two-wire system (direct current (dc) supply).
4 - Four-wire system (alternating current (ac) supply).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-C-5015 - Connectors, Electrical, Circular Threaded, AN Type, General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-P-24423 - Propulsion and Auxiliary Control Consoles and Associated Control and Instrumentation Equipment, Naval Shipboard Use, Basic Design Requirements.
MIL-M-38510 - Microcircuits, General Specification for.
MIL-C-52211 - Components and Assemblies for Industrial Gas Production, Storage and Transport Equipment, Packaging of.
MIL-P-55110 - Printed-Wiring Boards, General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

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MILITARY (Continued)

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MS3102 - Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type.
MS3106 - Connector, Plug, Electric, Straight, Solder Contacts, AN Type.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.

2.1.2 Other Government drawing. The following other Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
810-1385846 - Oxygen System, Submarine Replenishment.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

INSTRUMENT SOCIETY OF AMERICA (ISA)
Recommended Practice 37.1 - Nomenclature and Specification Terminology for Aerospace Test Transducers with Electrical Output.

(Application for copies should be addressed to the Instrument Society of America, P.O. Box 12277, Research Triangle Park, NC 27709.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of the specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The differential pressure transducers furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Definitions. Terminology consistent with Instrument Society of America (ISA) Recommended Practice 37.1 shall apply, as modified by the definitions specified in 6.4.

3.3 Description. The equipment specified herein shall be quantitative requirements needed to convert a fluid differential pressure to a continuous linear proportional analog electrical dc signal. The differential pressure transducer shall consist of the following components:

(a) Detector to sense the fluid differential pressure.
(b) Power supply to provide excitation energy to the detector and signal conditioner.
(c) Signal conditioner to convert the detector output to a continuous linear proportional analog electrical signal.

3.4 General requirements. Differential pressure transducer equipment shall be in accordance with provisions as specified herein.

3.4.1 Materials. Materials shall be in accordance with the prohibited and galvanic corrosion material requirements of MIL-E-917 and MIL-P-24423, and shall conform to requirement 4 of MIL-STD-454 for fungus inertness.

3.4.2 Sensing elements. The materials for the force summing devices and wetted parts on both the high and low pressure sides shall be in accordance with table I.

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TABLE I. Material.

Sensing element and wetted parts	Steam	Oil	Seawater	Service fluid flue gas and ammonia	Oxygen	Other gases, fresh water, and condensate
CRES 304L 316L and 321	X	X	—	X	—	X
CRES 17-4 PH 17-7 PH	X	X	—	—	—	X
Monel and K-Monel	X	X	X	—	X	X
Inconel 600 and X750	X	X	—	—	—	X
Inconel 625 and 718	X	X	X	—	—	X
Hastelloy C276	X	X	X	X	—	X
Titanium CP and 6Al-4V	X	X	X	—	—	X
CuNi 70/30	X	X	X	—	X	X
Ni span	X	X	—	—	—	X
Tantalum	X	X	X	—	—	X

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3.4.3 Ranges. Transducer differential pressure range shall be as specified (see 6.2.1). Ranges shall be in accordance with table II.

TABLE II. Ranges.

Type PDT	
Differential pressure ranges (PSID)	Differential pressure water column ranges (inches WCD)
0-15 0-30 0-60 0-100 0-200 0-400 0-600	0-10 0-60 0-150 0-300

3.4.4 Output. The electrical signal output of the transducer shall be dc directly proportional to differential pressure input. The output shall be a true current source of 4 to 20 milliamperes (mA) dc. The static error band requirement shall be met (see 3.7.1), regardless of external load resistance variations over a range of 0 to 250 ohms. The 4 mA output shall correspond to the lower differential pressure range value and the 20 mA output shall correspond to the upper differential pressure range value for the differential pressure ranges specified in table II.

3.5 Transducer equipment requirements. The transducer shall be constructed for an expected life of 40,000 hours of operation and subjected to the following conditions encountered in Naval service:

(a) High-impact shock (see 4.8.20).
(b) Vibration (see 4.8.19).
(c) Temperature (see 4.8.11) transducer ambient temperature operating range: 0 to 65 degrees Celsius (°C).
(d) Humidity (see 4.8.12).
(e) Supply voltage and frequency variations (see 4.8.7 and 4.8.10).
(f) Inclination (see 4.8.6).
(g) Differential pressure variations in the fluids from over range specified (see 4.8.17).
(h) Common mode pressure at pressure rating (see 4.8.8).

3.5.1 Calibration means. A means shall be provided at the signal conditioner to monitor transducer electrical output corresponding to differential pressure input to permit in-place calibration by one man. The signal from this means shall be of an accuracy sufficient to determine the static error band (see 3.7.1) and repeatability (see 3.7.2). The monitoring means may be in the form of test jacks. Monitoring shall be effected without the necessity of disconnecting the transducer from its source of electric power or altering its output loading.

3.5.2 Adjustments. Non-exposed adjustments for zero and span shall be provided for calibration purposes. The number of adjustments shall be kept to a minimum consistent with the operation and maintenance of the equipment.

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3.5.3 Enclosure. The differential pressure transducer shall be splash-proof as specified in MIL-STD-108.

3.5.4 Transducer mounting. The transducer shall have four mounting bolt holes, each 0.375 ± 0.005 inch in diameter. The centers of the mounting bolt holes shall form a rectangular pattern with sides of 3.125 ± 0.010 inches (lateral) and 5.375 ± 0.010 inches (longitudinal). The thickness of that portion of the mounting base through which the mounting bolts pass shall not exceed 0.50 inch.

3.5.5 External configuration. The transducer's external configuration shall conform to the following:

(a) The transducer's electrical connector (see 3.5.6) and differential pressure connections (see 3.5.7) shall be on opposite (longitudinal) ends of the transducer and shall face in opposite directions.
(b) The centers of the opening of the high and low pressure connections shall be within plus or minus 1.5 inches of the longitudinal centerline of the rectangular pattern formed by the four mounting-bolt holes. The lateral separation of the center of the opening of the high and low pressure connections shall not exceed 2.5 inches.
(c) The center of the face of the electrical connector shall lie within plus or minus 0.5 inch of the longitudinal centerline of the rectangular pattern formed by the four mounting-bolt holes.
(d) Exclusive of the pressure connections for submarine oxygen service (see 3.5.7), the end of the pressure connections shall be not less than 3.0 inches nor more than 5.5 inches from the geometric center of the pattern formed by the four mounting-bolt holes.
(e) The end of the electrical connector shall be not less than 3.0 inches nor more than 5.5 inches from the geometric center of the pattern formed by the four mounting-bolt holes.
(f) The center of the opening of the pressure ports shall be not less than 2.0 inches nor more than 3.75 inches from the underside of the mounting base.
(g) The center of the face of the electrical connector shall be not less than 0.75 inch nor more than 3.0 inches from the underside of the mounting base.
(h) The total height shall not exceed 6.25 inches.
(i) The total width shall not exceed 4.20 inches and neither side shall extend more than 2.10 inches from the geometric center of the pattern formed by the four mounting-bolt holes.

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(j) Exclusive of the pressure connections for submarine oxygen service (see 3.5.7), the total length (including the pressure connection and electrical connector) shall not exceed 9.0 inches.

(k) Without having to unmount the transducer, the zero adjustment, span adjustment, and calibration test points shall be accessible when transducers are installed side-by-side with lateral separations of 0.25 inch or less.

3.5.6 Electrical connector. An electrical interface connector (receptacle and mating plug) in accordance with MIL-C-5015 shall be provided with each transducer.

3.5.6.1 Transducers 28-volt dc. The receptacle shall be an MS3102R14S-5P in accordance with MS3102. Receptacle pin "A" shall be positive, pin "B" shall be negative, and pin "C" shall be case ground. The mating plug shall be an MS3106F14S-5S in accordance with MS3106.

3.5.6.2 Transducers 115-volt ac. The receptacle shall be a MS3102R14S-5PX in accordance with MS3102. Receptacle pins "A" and "B" shall be 115 volts ac input, pin "C" shall be case ground, pin "D" shall be positive 4 to 20 mA output, and pin "E" shall be negative 4 to 20 mA output. The mating plug shall be an MS3106F14S-5SX in accordance with MS3106.

3.5.7 Differential pressure connections. Transducer differential pressure sensing connections for all services shall be in accordance with MS16142. Only 1/4-inch outside diameter tube size shall be used. However, for use on submarine oxygen replenishment systems shown on Drawing 810-1385846, the transducer sensing connection shall be a nickel-copper pipe nipple 1/4-inch nominal pipe size with 0.120-inch minimum wall thickness, 6 inches long, welded to the socket in accordance with MIL-STD-278 as specified (see 6.2.1). The high and low pressure connections shall be clearly labeled by marking on the detector body adjacent to the connections. Facing the pressure inlet ports and with the mounting surface horizontal downward, the high pressure port shall be on the left and the low pressure port shall be on the right.

3.5.8 Power supply requirements. Nominal steady-state power supply requirements for ac shall be 115 $\pm$ 8 volts, 60 $\pm$ 2 hertz (Hz), single-phase. Nominal steady-state power supply requirements for dc shall be 28 $\pm$ 4.5 volts. The transducer shall operate with power supply variations as specified in 4.8.7 and 4.8.10.

3.5.9 Detector cleaning. Prior to packaging for delivery, the contractor shall ensure that differential pressure transducers shall be free of all loose scale, rust, grit, filings and other foreign substances and free of mercury, oil, grease or other organic materials. Oil may be present only in detectors for SW, OS, and FG applications. The surfaces to be cleaned shall be the pressure connection and all parts contacted by the measurement fluid, except for oxygen, where "clean" shall include all surfaces of the detector. In addition, the following shall apply:

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(a) Detector for OX service shall be clean gas calibrated, cleaned, and packaged in accordance with MIL-C-52211.
(b) Detector for GW service shall be clean gas calibrated and pressure connections capped.
(c) Detector for ST service shall be freshwater or clean gas calibrated and pressure connections capped.
(d) Detector for OS, SW, and FG service shall be oil, water, or clean gas calibrated and pressure connections capped.

3.5.9.1 Special marking. Transducer for use with OX and GW service shall have "USE NO OIL FOR CALIBRATION" prominently marked on the detector.

3.5.10 Welding and bolting. For OX service, all joints in the pressure system shall be welded or bolted.

3.5.11 Lubrication. The transducer shall operate without lubrication of any moving part after assembly. The legend "DO NOT LUBRICATE" shall be prominently marked on the detector.

3.5.12 Damping. The use of oil for damping, as a fill media, and so forth, in transducers for OX and GW applications shall be cited on the equipment drawing.

3.5.13 Weight. The weight of a differential pressure transducer, including mounting brackets, shall not exceed 20 pounds.

3.6 Detail requirements.

3.6.1 Fuses. The transducer shall not be fused.

3.6.2 Semiconductors. Silicon rectifiers, diodes, transistors and other semiconductor devices shall be in accordance with MIL-S-19500 as applicable, and shall be selected as specified in MIL-STD-701. Devices other than those listed in MIL-STD-701 shall be used only where specifically approved in writing by NAVSEA on the basis of justification submitted in accordance with MIL-M-38510.

3.6.3 Electron tubes. Electron tubes shall not be used.

3.6.4 Modular assemblies. The power supplies, signal conditioners, and other circuitry separate from the detector shall meet the modular construction requirements specified in 3.6.4.1 and 3.6.4.2.

3.6.4.1 Plug-in units. The plug-in module units may include, but are not limited to, printed circuit boards for conventional components, cable connectors, power supplies, and integrated microelectronic circuits.

3.6.4.2 Accessibility. Modular units or assemblies shall be fastened in a manner to allow for removal for maintenance, accessibility, or replacement.

3.6.5 Printed wiring boards. Printed wiring boards shall be constructed in accordance with MIL-STD-275 and MIL-P-55110.

3.6.6 Resistors. Composition type fixed or variable resistors shall not be used.

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3.7 Transducer performance. Unless otherwise specified herein, performance tolerances are specified in percent of transducer span.

3.7.1 Static error band. The transducer static error band shall not exceed plus or minus 1 percent (see 4.8.1).

3.7.2 Repeatability. Repeatability of transducer output shall be within 0.5 percent (see 4.8.1).

3.7.3 Sensitivity factor. The values of the ratio of electrical ouput percentage change to pressure input percentage change shall be not less than 0.80 or more than 1.20 (see 4.8.3).

3.7.4 Ripple. Transducer root mean square (rms) output ripple shall not exceed 0.15 percent of full scale output (see 4.8.4).

3.7.5 Warm-up time. The transducer output shall attain a value within plus or minus 1 percent of the steady-state output with no overshoot in excess of 1 percent. Output shall reach this band in 15 minutes or less after the transducer is energized and shall remain in this band (see 4.8.5).

3.7.6 Inclination. Maximum deviation of the transducer output resulting from inclination shall not exceed 3.0 percent for transducers with upper range values exceeding 60 inches of water column differential, and 4.0 percent for transducers with upper range values of 60 inches of water column differential and below (see 4.8.6).

3.7.7 Supply voltage and frequency or supply voltage (for dc) (steady-state). Maximum difference between outputs at any voltage and frequency or voltage (for dc) condition and the normal (115 volts, 60 Hz or 28 volts dc (Vdc)) at the same input, differential pressure, and test temperature shall not exceed 1 percent (see 4.8.7).

3.7.8 Common mode pressure. During the common mode pressure test, transducer performance shall be within the range formed by extending the upper and lower static error band limits specified in 3.7.1 by an amount equal to the following (see 4.8.8):

$$(1/10) \quad \left(\frac{\text{System operating pressure}}{\text{D/P range}}\right)^{1/3}$$

3.7.9 Response (see 4.8.9). Transducer output shall conform to the following criteria, where all percentages are of transducer span:

(a) Transducer output shall be within plus or minus 2 percent of maximum ramp pressure within 0.01 second of the time that pressure is attained.
(b) Transducer output shall exhibit no overshoot of maximum ramp pressure in excess of 2.0 percent.
(c) Transducer output shall indicate the actual pressure to within plus or minus 1.0 percent in 0.15 second or less after attainment of maximum ramp pressure, and shall remain within this error band for the duration of applied steady-state pressure.

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3.7.10 Supply voltage and frequency (transient) (see 4.8.10).

3.7.10.1 Voltage. During the voltage transient test, the transducer output shall remain within plus or minus 5 percent of the pre-transient output.

3.7.10.2 Frequency. During the frequency transient test, the transducer output shall remain within plus or minus 0.5 percent of the steady-state output.

3.7.11 Temperature. During the temperature test, transducer performance shall be within the static error band specified in 3.7.1 (see 4.8.11).

3.7.12 Humidity. Maximum deviation of the second cycle reference measurement from the post-conditioning calibration shall not exceed 1.5 percent. After temperature cycling, performance shall conform to 3.7.1 and shall show no evidence of damage (see 4.8.12).

3.7.13 Overpressure. Calibration conducted after overpressure test shall have no values in excess of 1 percent deviation from the pre-overpressure test calibration. After calibration, the transducer shall adjust to the accuracy requirements of 3.7.1 (see 4.8.13).

3.7.14 Line pressure. After the line pressure test, transducer performance shall be within the static error band specified in 3.7.1 (see 4.8.14).

3.7.15 Salt spray. No appreciable corrosion or other damage, either mechanical or electrical, shall be evident after the salt spray test. After completion of the test, performance shall conform to 3.7.1 (see 4.8.16).

3.7.16 Differential pressure whip. Calibration conducted after completion of differential pressure whip test shall have no values in excess of 1 percent deviation from pretest reference measurement (see 4.8.17).

3.7.17 Insulation resistance. The insulation resistance of the transducer shall be not less than 10 megohms (see 4.8.18).

3.7.18 Vibration. Monitored transducer output during all phases of vibration test shall show no variation from steady-state output in excess of 2 percent. There shall be no visible evidence of damage to the transducer as a result of the vibration test (see 4.8.19).

3.7.19 Shock. After the shock test, but before any adjustment is made, the transducer output shall show no deviation greater than 3 percent for transducers with upper range values exceeding 60 inches of water column differential, and 4 percent for transducers with upper range values of 60 inches of water column differential and below. A post-test calibration, using adjustments provided, shall conform to performance requirements of 3.7.1. There shall be no visual evidence of damage to the transducer as a result of the shock test (see 4.8.20).

3.7.20 Burst pressure. The transducer shall withstand the burst pressure specified in 4.8.21 without showing any evidence of leakage.

3.7.21 Output. The output shall conform to 3.4.4 and the transducer performance shall be within the static error band specified in 3.7.1 (see 4.8.22).

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3.7.22 Enclosure. There shall be no evidence of water leakage into the equipment enclosure (see 4.8.15).

3.7.23 Electromagnetic interference emission and susceptibility (EMI). The differential pressure transducers shall meet the following requirements of MIL-STD-461: CE01, CE03, CS01, CS02, CS06, CS09, RE01, RS01, RS02, RS03, except as modified below (see 4.8.23):

(a) Requirement CS02 shall be met as specified, except as follows:

(1) At frequencies between 2 and 30 megahertz, the full output of the generator (source impedence of 50 ohms, and an output capability of 12.25 volts rms, or 3 watts, when connected to a 50-ohm load) shall be applied to all power and signal leads of the test sample.
(2) The "calibrating resistor", or 50-ohm load, shown in the CS02 test block diagram of MIL-STD-462, shall not be connected during the test.
(3) The test sample output shall not vary by more than 1.0 percent of transducer span.
(4) The test signal shall be applied to all power and signal leads of the test sample.

(b) Requirement RS03 shall be met as specified, except as follows:

(1) Between the frequencies of 14 kilohertz and 18 gigahertz, the electric field strength test level shall be 10 volts per meter.

(c) Requirement CS06 shall be met as specified, except as follows:

(1) The test signal shall be applied only to ac power leads of the test sample.

(d) Requirements CE01 and CE03 shall be met as specified, except as follows:

(1) The test signal shall be applied only to ac power leads of the test sample.

3.8 Workmanship. Workmanship of differential pressure transducers shall conform to requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order, an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Periodic inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. Qualification inspection shall consist of the examination and tests specified in table III.

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TABLE III. Qualification inspection.

Examination and tests	Requirement	Examination and test
General examination	3.4	4.6
Output	3.7.21	4.8.22
Warm-up time	3.7.5	4.8.5
Enclosure	3.7.22	4.8.15
Salt spray	3.7.15	4.8.16
Static error band and repeatability	3.7.1 and 3.7.2	4.8.1
Sensitivity factor	3.7.3	4.8.3
Ripple	3.7.4	4.8.4
Inclination	3.7.6	4.8.6
Supply voltage and frequency (steady-state)	3.7.7	4.8.7
Common mode pressure	3.7.8	4.8.8
Response	3.7.9	4.8.9
Supply voltage and frequency (transient)	3.7.10	4.8.10
Temperature	3.7.11	4.8.11
Humidity	3.7.12	4.8.12
Overpressure	3.7.13	4.8.13
Line pressure	3.7.14	4.8.14
Differential pressure whip	3.7.16	4.8.17
Insulation resistance	3.7.17	4.8.18
Vibration	3.7.18	4.8.19
Shock	3.7.19	4.8.20
Burst pressure	3.7.20	4.8.21
Electromagnetic interference emission and susceptibility	3.7.23	4.8.23

4.3.1 Samples for qualification. The number of samples to be submitted for the qualification tests specified in table III shall depend on the transducer design. If each range is covered by a separate and distinct design, a sample for each range will require testing. In instances where a singular design series may cover several ranges and types, only one sample need be submitted for testing, provided the electrical and mechanical similarities are approved by NAVSEA. In no case, however, shall less than three units, one unit each representing low (300 WCD and below), mid (15 to 99 PSID), and high (100 PSID and above) ranges, be submitted for qualification testing, regardless of design similarity. Successful passage of the qualification tests by each representative sample shall qualify that design for that discrete range; that is, low, mid, or high.

4.3.2 Authorization. Prior to authorization of qualification tests, the contractor shall submit complete mechanical and electrical drawings to NAVSEA for review and acceptance.

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4.3.3 Retention of qualification. To retain qualification, the contractor shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, indicating as a minimum the number of lots that have passed, the number that have failed, and the group which they failed. The results of tests of all reworked lots shall be identified and accounted for.

(b) A summary of the results of tests performed for periodic inspection, including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the contractor may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements and the reason for no production.

4.4 Quality conformance inspection. Each differential pressure transducer in each lot offered for delivery shall be subjected to the inspection listed in table IV and shall be conducted in the order listed. Failure of any differential pressure transducer to meet the requirements of this specification shall be cause for rejection. An inspection lot shall consist of all differential pressure transducers of the same classification (see 1.2) produced under the same conditions and offered for delivery at the same time.

TABLE IV. Quality conformance inspection.

Inspection	Requirement	Examination and test
General examination	3.4	4.6
Output	3.7.21	4.8.22
Static error band and repeatability	3.7.1 and 3.7.2	4.8.1
Overpressure	3.7.13	4.8.13
Line pressure	3.7.14	4.8.14
Insulation resistance	3.7.17	4.8.18

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4.5 Periodic inspections. Three transducers shall be selected at random out of every 100 transducers which have satisfactorily passed the tests in 4.4 from each lot offered for delivery, and shall be subjected to the tests listed in table V. If any transducer fails in any test, no transducers shall be accepted for quality conformance inspection until the contractor has determined the cause of the defect and has taken the necessary action to correct or eliminate the defects from all units on hand. The failed test and any other periodic tests required by the Government shall be repeated to demonstrate that the corrective action shall enable the transducer to conform to the requirements of this specification. All design changes results of all repeated tests require NAVSEA approval and certification, respectively. Samples that have satisfactorily passed the examination and tests listed in table V may not be applied as part of the quantity specified (see 6.2.1).

TABLE V. Periodic inspections.

Test	Requirement	Test
Inclination	3.7.6	4.8.6
Supply voltage and frequency (steady state)	3.7.7	4.8.7
Common mode pressure	3.7.8	4.8.8
Temperature	3.7.11	4.8.11
Burst pressure	3.7.20	4.8.21

4.6 General examination. Each transducer shall be given a thorough examination to determine conformance to the requirements of this specification with respect to material, finish, workmanship, construction, assembly, dimensions, weight, marking of identification, and information plates. Examination shall be limited to the examinations that may be performed without disassembling the units in such a manner that its performance, durability, or appearance would be affected. Examination shall also include a check of all operation controls and adjustments, as applicable.

4.7 Test conditions and test procedures.

4.7.1 Test conditions. Except where the following factors are the variables, the tests specified in 4.8 shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature shall be 25 ± 2°C.
(b) Relative humidity shall be ambient.
(c) Supply voltage shall be 115 volts (nominal) for power supply designation 4 or 28 volts (nominal) for power supply designation 2.
(d) Supply frequency shall be 60 Hz (nominal) for power supply designation 4 or direct current for power supply designation 2.
(e) Except where the external load is a test requirement, any value of resistance between 0 and 250 ohms may be employed.

4.7.2 Calibration system requirements. When specified in the contract or order, equipment calibration procedures shall be prepared (see 6.2.2).

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4.8 Tests. Except for the warm-up time test (see 4.8.5), the transducer and all associated test equipment shall be energized for a period of time sufficient to ensure complete warm-up.

4.8.1 Static error band and repeatability. The transducer shall first be flexed over its full pressure range by slowly increasing and decreasing the applied pressure for six continuous cycles. The calibration measurement shall be made at a minimum of five equally spaced intervals over the entire range (both upscale and downscale). Precaution shall be taken to avoid overshoot. This calibration procedure shall be applied three successive times to determine repeatability. Static error band of all calibration shall meet the requirements of 3.7.1. Repeatability shall meet the requirements of 3.7.2.

4.8.2 Reference measurement. The following reference measurement shall be conducted when specified in the individual test:

(a) A one-trial calibration with at least five equally spaced intervals over the entire transducer range both upscale and downscale.

4.8.3 Sensitivity factor. The sensitivity factor shall be determined as follows. Provide a differential pressure to the transducer to a value of 80 ± 5 percent of span. Record the input differential pressure and corresponding electrical output. Increase the differential pressure by an amount not exceeding 0.25 percent of span. Record both the new differential pressure and corresponding new electrical output. Calculate the change in both applied differential pressure and electrical output as a percentage of transducer span. Determine the ratio of the electrical output percentage change to applied differential pressure percentage change. Repeat this procedure for a differential pressure decrease not exceeding 0.25 percent of span. Performance shall conform to the requirements of 3.7.3.

4.8.4 Ripple. Rms ripple shall be determined at an input differential pressure of 80 ± 5 percent of transducer span. Performance shall conform to the requirements of 3.7.4.

4.8.5 Determination of warm-up time. The test specified in 4.8.5.1 shall be conducted to determine the elapsed time between the application of line power to the transducer and the point at which the transducer output reaches the conditions specified in 3.7.5.

4.8.5.1 Test conditions. The transducer shall be placed in an ambient temperature of 25 ± 2°C for not less than 2 hours de-energized. Recording equipment and other auxiliary equipment shall be energized to assure complete warm-up. An input differential pressure of 80 ± 5 percent of the transducer span shall be applied to the transducer and maintained constant during this test. Performance shall conform to 3.7.5.

4.8.6 Inclination. Prior to any inclination, a reference measurement in accordance with 4.8.2 shall be made with the transducer in each of the following orientations:

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(a) Mounting surface horizontal.
(b) Mounting surface vertical with pressure connections vertical downward.

From each of the mounting orientations specified above, the transducer shall be inclined for at least 1 minute in each of the following positions and a reference measurement in accordance with 4.8.2 shall be made. Performance shall conform to 3.7.6.

(a) 45 degrees from the vertical forward.
(b) 45 degrees from the vertical backward.
(c) 45 degrees from the vertical to the left.
(d) 45 degrees from the vertical to the right.

4.8.7 Supply voltage and frequency or supply voltage (for dc) (steady-state). The transducer shall be operated at the configurations of normal, maximum and minimum steady-state voltages. The ambient temperature shall also vary, with the transducer operated for at least 15 minutes at each test temperature before the first reference measurement. The transducer shall be allowed at least 15 minutes to stabilize at each configuration. After that, a reference measurement shall be taken (see 4.8.2). Reference measurements shall be performed at ambient temperatures of 0 ± 2°C, 25 ± 2°C and 65 ± 2°C. Test temperatures shall be accomplished by varying temperature in steps of 10°C each (30 minutes for each step) until the desired ambient temperature is reached. Performance shall conform to 3.7.7.

4.8.8 Common mode pressure test. The transducer's rated pressure specified (see 6.2.1) shall be applied simultaneously to both pressure ports. The pressure at the low pressure port shall then be decreased (in pressure increments specified in 4.8.2) to the specified transducer range, and then increased in similar increments to the transducer rated pressure. Performance shall conform to 3.7.8.

4.8.9 Response. A differential pressure ramp consisting of a differential pressure rise of at least 40 percent of transducer span occurring at a rate of not less than 400 percent per second shall be applied to the transducer. The maximum ramp pressure shall be maintained for at least 0.75 second and shall not vary by more than plus or minus 2 percent of transducer span. Performance shall conform to 3.7.9.

4.8.10 Supply voltage and frequency or supply voltage (for dc) (transient). Tests specified in 4.8.10.1 and 4.8.10.2 shall be conducted with a differential pressure input signal equal to 80 ± 5 percent of the transducer span. Performance shall conform to the requirements of 3.7.10.

4.8.10.1 Transient voltage.

4.8.10.1.1 Upper limit of steady-state voltage. With the transducer operating at the upper limit of steady-state ac voltage, the ac-powered transducer shall have a transient voltage of plus 15 volts recovering to the steady-state band in 2 seconds superimposed. With the transducer operating at the upper limit of steady-state dc voltage, the dc-powered transducer shall have a transient voltage of plus 2.0 volts recovering to the steady-state band in 2 seconds superimposed.

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4.8.10.1.2 Lower limit of steady-state voltage. With the transducer operating at the lower limit of steady-state ac voltage, the ac-powered transducer shall have a transient voltage of minus 15 volts recovering to the steady-state band in 2 seconds superimposed. With the transducer operating at the lower limit of steady-state dc voltage, the dc-powered transducers shall have a transient voltage of minus 2.0 volts recovering to the steady-state band in 2 seconds superimposed.

4.8.10.2 Transient frequency (for ac-powered transducers).

4.8.10.2.1 Upper limit of steady-state frequency. With the transducer operating at the upper limit of steady-state frequency, a transient frequency of plus 1.5 Hz recovering to the steady-state band in 2 seconds shall be superimposed.

4.8.10.2.2 Lower limit of steady-state frequency. With the transducer operating at the lower limit of steady-state frequency, a transient frequency of of minus 1.5 Hz recovering to the steady-state band in 2 seconds shall be superimposed.

4.8.11 Temperature. The transducer shall operate normally (without alignment or adjustment) throughout the following temperature cycle. Tolerances in operating characteristics shall be as specified herein:

(a) Hold test temperature at 0 ± 2°C for at least 24 hours.
(b) Increase test temperature in steps of 10°C each, at 30 minutes for each step, until plus 65 ± 2°C is reached and hold at that temperature for at least 4 hours.
(c) Reduce test temperature in steps of 10 degrees each, at 30 minutes for each step, until plus 25 ± 2°C is reached and hold at that temperature for at least 4 hours.

At each temperature plateau (0, 65, and 25°C), a reference measurement (see 4.8.2) shall be made. Performance shall conform to 3.7.11.

4.8.12 Humidity. Transducer shall be subjected to the following conditioning and test. Except for the periods of test indicated below, the equipment shall not be energized. Performance shall meet the requirements of 3.7.12.

4.8.12.1 Conditioning. In order to establish a reference condition for the measurement of operating parameters and a valid basis for comparison of the effects of the conditioning to follow, the complete equipment shall be dried at a temperature not less than 40°C or more than 50°C for at least 2 hours.

4.8.12.2 Reference measurements. Following the conditioning specified in 4.8.12.1, a reference measurement in accordance with 4.8.2 shall be conducted at 25 ± 5°C and 50 ± 5 percent relative humidity to indicate satisfactory performance.

4.8.12.3 Temperature cycling. The transducer shall then be subjected to four 24-hour cycles of temperature variation consisting of approximately 18 hours at 65 ± 5°C and approximately 6 hours at 25 ± 5°C. The relative humidity shall be maintained above a minimum of 95 percent during the steady-state conditions.

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The transitions between temperatures shall be accomplished within the 6-hour period so that the time at the high temperature is approximately 18 hours. Each transition shall not exceed 1 hour if the transducer remains in the chamber, or 15 minutes if a two-chamber method is employed. The relative humidity need not be controlled during the transition periods.

4.8.12.4 Measurement during cycling. During the second cycle, the measurement required in 4.8.12.2 shall be performed at 65 ± 5°C before the temperature is decreased to 25°C with the transducer remaining in the chamber. The transducer shall be energized for as brief a period as required to complete the measurements. A warm-up period is permitted where previous tests indicate a definite period is required for the transducer to attain thermal stability.

4.8.12.4.1 Requirements of transducer. The transducer shall meet the requirements specified herein without alignment or adjustments, other than the accessible controls employed for operation of the equipment. No repairs are permitted before measurements are taken. If repairs are required, the test specified in 4.8.12.3 shall be repeated, after the necessary replacements have been made.

4.8.12.5 Measurements after temperature cycling. After the four complete cycles, the measurements required in 4.8.12.2 shall be performed at 25 ± 5°C with the transducer remaining in the chamber. The transducer shall meet the requirements of this specification with no readjustments.

4.8.12.5.1 Additional tests. Upon completion of the tests, and after remaining inoperative for not less than 12 hours nor more than 24 hours at a temperature of 25 ± 5°C and 50 ± 5 percent relative humidity, any additional tests or measurements considered necessary by the Government shall be made to determine conformance to the requirements specified herein. In addition, the transducer shall be examined to detect evidence of physical degradation such as corrosion of metal parts or distortion of plastic parts. When it is necessary to replace parts to obtain satisfactory transducer performance, the failed part or parts shall be analyzed to determine the cause of unsatisfactory operation. The results of the analysis shall be recorded with the results of measurements of the transducer's operating parameters. Unsatisfactory parts or materials shall be replaced by adequate substitutes.

4.8.13 Overpressure. Before the overpressure test, a reference measurement in accordance with 4.8.2 shall be made. The transducer shall successfully withstand differential pressure equal to 200 percent of its range with a maximum pressure of 10,000 lb/in^2 for a period of 1/2 hour. At the end of this period, transducers shall be immediately subjected to a pressure equal to 1 lb/in^2 or 10 percent of range, whichever is less, below atmospheric for an additional period of 1/2 hour. Within 10 minutes after release of this differential pressure, a reference measurement (see 4.8.2) shall be made for comparison. Performance shall conform to 3.7.13. When the line pressure rating exceeds 200 percent of the differential pressure range, the overpressure test shall be omitted and the 1 percent deviation allowed in 3.7.13 shall be applied to the line pressure test.

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4.8.14 Line pressure. The transducer shall successfully withstand the pressure rating specified (see 6.2.1) when applied to the high pressure port with the low pressure port vented to atmosphere for a period of 10 minutes. The above shall be repeated with the pressure applied to the low pressure port of the transducer. After each test, a reference measurement in accordance with 4.8.2 shall be made. Performance shall conform to 3.7.14.

4.8.15 Enclosure. The enclosure shall be subjected to the splashproof test specified in MIL-STD-108. Performance shall conform to 3.7.22.

4.8.16 Salt spray. The salt spray test shall be conducted in accordance with method 101D of MIL-STD-202 using a 20 percent salt solution. Duration of the test shall be 100 hours. The internals of the signal conditioner and power supply shall not be subjected to the test. When any transducer component is physically attached to the detector, the integral component shall also be tested. If the transducer is not designed for seawater service, the inlet pressure connections shall be plugged. Performance shall conform to 3.7.15.

4.8.17 Differential pressure whip. Before performing the differential pressure whip test, a reference measurement shall be made (see 4.8.2). The test shall be conducted on a suitable system by applying a sinusoidal pressure change of not more than 20 percent to not less than 80 percent of span for a total of 260,000 cycles. The rate of cycling shall be approximately 1 Hz. The transducer shall be energized throughout the test. After completion of the differential pressure whip test, a reference measurement shall be made for comparison (see 4.8.2). Performance shall conform to 3.7.16.

4.8.18 Insulation resistance. The insulation resistance of the transducer shall be determined by applying 500 Vdc between electrical input and output circuits and between these circuits and ground. The temperature shall be 25 $\pm$ 5°C and the relative humidity shall be 50 $\pm$ 10 percent. The insulation resistance measurement shall be made immediately after a 2-minute period of uninterrupted test voltage application. However, if the indication of insulation resistance meets the specified limit (see 3.7.17), and is steady or increasing, the test may be terminated before the end of the 2-minute period.

4.8.19 Vibration. The transducer shall be tested in accordance with type I (environmental) vibration of MIL-STD-167-1, except that the upper frequency shall be 175 Hz, the amplitude of vibration shall be in accordance with table VI, and for the variable frequency portion, the vibration level shall be maintained for 2 minutes at each integral value of frequency. If no resonances are observed, the 2-hour endurance test shall be conducted at 175 Hz. During the vibration test, a differential fluid pressure of 80 $\pm$ 5 percent of the transducer span shall be applied to the transducer. Transducer output during the test shall be monitored. Performance shall conform to 3.7.18.

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TABLE VI. Amplitudes of vibration.

Frequency range (Hz)	Table amplitude (inch)	
	Exploratory	Variable frequency
5 to 20	0.010 ± 0.002	0.030 ± 0.006
21 to 50	0.004 ± 0.001	0.020 ± 0.004
51 to 100	0.0015 ± 0.0003	0.010 ± 0.002
101 to 175	0.0005 ± 0.0001	0.0015 ± 0.0003

4.8.20 Shock. The shock tests shall be conducted in accordance with grade A, class 1, type C using bulkhead mounting fixture 4-A of MIL-S-901. During the test, a fluid differential pressure input equal to 80 ± 5 percent of the transducer span shall be applied to the transducer. Transducer output during the test shall be monitored. Before and after the shock test, reference measurements shall be made for comparison (see 4.8.2). Performance shall conform to 3.7.19.

4.8.21 Burst pressure. The transducer shall be subjected to a liquid pressure equal to 300 percent of the range, with a maximum pressure of 10,000 lb/in^2 applied simultaneously to both sides of the transducer for a period of 10 minutes. The transducer shall conform to the requirements of 3.7.20. No performance tests shall be required after the burst pressure test. A reference measurement (see 4.8.2) shall be recorded for information purposes.

4.8.22 Output. A reference measurement shall be made in accordance with 4.8.2. Performance shall conform to the requirements of 3.7.21.

4.8.23 EMI tests. EMI tests shall be in accordance with the test methods specified in MIL-STD-462, with the modifications as specified in 3.7.23. Performance shall be as specified in 3.7.23.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

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5.1 Packaging requirements. Transducer equipment shall be individually preserved level A, C, or commercial, packed level A, B, C, or commercial as specified (see 6.2.1), marked (including bar coding) in accordance with MIL-E-17555, and shall include acquisition options therein as specified (see 6.2.1). The transducer components shall be packaged as a single unit. Unless otherwise specified (see 6.2.1), method III shall apply for level A preservation. In addition, for Navy acquisitions, the following applies:

(a) Navy fire-retardant requirements.

(1) Lumber and plywood. Unless otherwise specified (see 6.2.1), all lumber and plywood (including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing) shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes (including interior packaging forms) shall meet the flamespread and the specific optic density requirements of PPP-F-320.

6. NOTES

6.1 Intended use. Transducers are intended to provide control input and monitoring of differential pressures for shipboard engineering plants.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Classification designations required (see 1.2.1).
(c) Pressure rating (see 1.2.1.7 and 4.8.8).
(d) Transducer differential pressure range (see 3.4.3 and 4.8.14).
(e) Type of differential pressure connections (see 3.5.7).
(f) Quantity of transducer required (see 4.5).
(g) Level of preservation and packing required (see 5.1).
(h) Applicable acquisition requirements of MIL-E-17555 (see 5.1).
(i) Requirements for level A preservation, if other than specified (see 5.1).
(j) When fire retardant material is not required (see 5.1(a)(1)).
(k) Flamespread and specific optic density requirements, if other than as specified (see 5.1(a)(2)).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	Inspection system program plan	DI-R-4803	----
4.7.2	Equipment calibration procedures	DI-R-7065	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, Military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be exhibited on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24304 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

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6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Definitions.

6.4.1 Static error band. Static error band is the maximum deviation from a straight line drawn through the coordinates of the lower range limit at specified transducer output, and the upper range limit at specified transducer output expressed in percent of transducer span.

6.4.2 Common mode pressure. The common mode pressure is static line pressure applied simultaneously to both pressure sides of the transducer.

6.4.3 Overpressure. Overpressure is the maximum expected differential pressure, above or below zero, expressed in percent of differential span of unit.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 NATO coordination and standardization. The standard characteristics of the differential pressure transducer supplied for U.S. Navy ships have been coordinated with NATO standardization document, where applicable. In particular, the standard characteristics of the differential pressure transducer output conforms to corresponding transducer output specified in STANAG 4286.

6.8 Subject term (key word) listing.

Current
Fluid
Range
Sensing

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6685-N790)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-24304B(SH)	DIFFERENTIAL PRESSURE TRANSDUCER EQUIPMENT (ELECTRICAL) (NAVAL SHIPBOARD USE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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